MWALIMU EXAMINATIONS BUREAU

UACE RESOURCE MOCK EXAMINATIONS 2019

P525/2 CHEMISTRY PAPER 2 GUIDE

1. (a) (i) Electropositivity increases from beryllium to barium (down the group). Down the group, nuclear charge/attract ion increases as protons are added to the nucleus. The screening effect also increases due to the extra full energy level (shell) of electrons added. The increase ions screening effect is more than the increase in nuclear charge, therefore net (effective) nuclear decreases. Hence the nuclear attraction for the outermost electrons is less. This causes an increase in electropositivity. **(03 Mks)**

(ii) First ionization energy decreases from beryllium to barium. Down the group, atomic radius increases. The nuclear attraction for the outermost electrons reduces due to reduced effective nuclear charge. This makes the outermost electrons easily lost and hence first ionization energy decreases. **(03 Mks)**

(b) (i) Beryllium does not react with water at all conditions. Hot magnesium readily burns in steam to form magnesium oxide and hydrogen. **(03½ Mks)**

$Mg_{(s)} + H_2O_{(g)} \longrightarrow MgO_{(s)} + H_{2\,(g)}$

Calcium strontium and barium react with cold water to liberate hydrogen gas and a corresponding hydroxide

$X_{(s)} + H_2O_{(l)} \longrightarrow X(OH)_{2(s/aq)} + H_2$; X= Ca, Sr, or Ba

(ii) All react with dilute hydrochloric acid at room temperature except beryllium that requires warming to the corresponding chloride and hydrogen gas is evolved. **(03 Mks)**

$X_{(s)} + 2H^+_{(aq)} \longrightarrow X^{2+}_{(aq)} + H_{2(g)}$; X = Be, Mg, Ca, Sr or Ba

(iii) All react with dry chlorine on (strong) heating to form the corresponding chlorides

$M_{(s)} + Cl_{2\,(g)} \longrightarrow MCl_{2\,(s)}$; M = Be, Mg, Ca, Sr, or Ba **(02½ Mks)**

(c) (i) Both are rendered passive by concentrated nitric acid **(04 Mks)**

- Both react with hot concentrated (caustic) alkalis to evolve hydrogen gas and complex ions are left in solution.
- Their oxides (and hydroxides) are amphoteric.
- Their chlorides are covalent (or are soluble in organic solvents) (and readily hydrolyze in water to form an acidic solution).
- Their carbides readily hydrolyze in water to evolve methane gas (and a corresponding hydroxide is formed). *-01 for extra wrong/correct*

(ii) -Their atoms have the same electronegativity/electropositivity **(01 Mk)**

- Their atoms have similar (almost the same) electrode potential
- Their cations have similar charge density/polarizing power. *-0½ for extra correct/wrong*

2. (a) (Nuclear) fusion is a process where light nuclei combine to form heavier nuclei with release of energy whereas (nuclear) fusion is the braking/splitting of a heavy nuclide into smaller nuclides with release of energy.

(02 Mks)

(b)(i) See graph paper; plotting axes (labeled correctly and scale, shape (smooth curve)

(03 Mks)

(ii)

Mass (g)	half life(s)
60 to 30	64
30 to 15	68

$t_{1/2} = (64 + 68) \div 2 = 66s$ **(01 Mk)**

(ii) $t_{1/2} = \frac{0.693}{\lambda} \Rightarrow \lambda = \frac{0.693}{66} = 1.05 \times 10^{-2} s^{-1}$ **(02 Mks)**

(iv) $\log\left(\frac{a_o}{a_t}\right) = \frac{\lambda t}{2.303} \Rightarrow \log\left(\frac{a_o}{a_t}\right) = \frac{1.05 \times 10^{-2} \times 198}{2.303}$ **(03 Mks)**

$$\frac{a_o}{a_t} = 10^{0.90274}$$

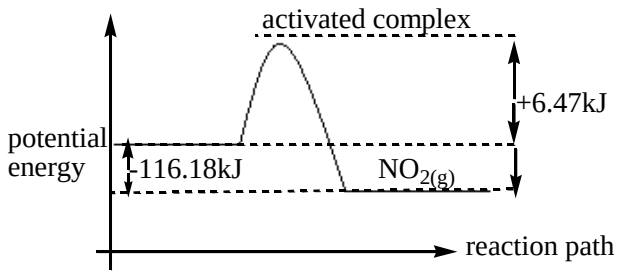
$$\frac{a_t}{a_o} \times 100 = \left(\frac{1}{10^{0.90274}} \times 100\right) = 12.51\%$$

(c) (i) Rate reduces to ¼ times the original **(01 Mk)**

(ii) Rate increases to 8 times the original **(01 Mk)**

(d) $K = \frac{108}{(0.64)^2(0.021)} = 12{,}555.8\ N^{-2}M^{4}S^{-1}$ **(02 Mks)**

(e) (i)

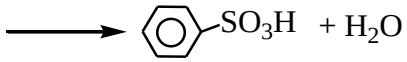


shape + lebels **(03 Mks)**

(ii) Activation energy reduces. Catalyst provides an <u>alternative reaction path with much lower activation energy</u>

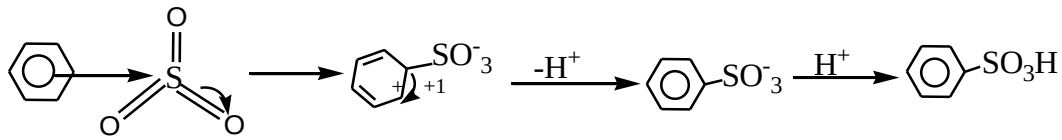
(02 Mks)

3. (a)

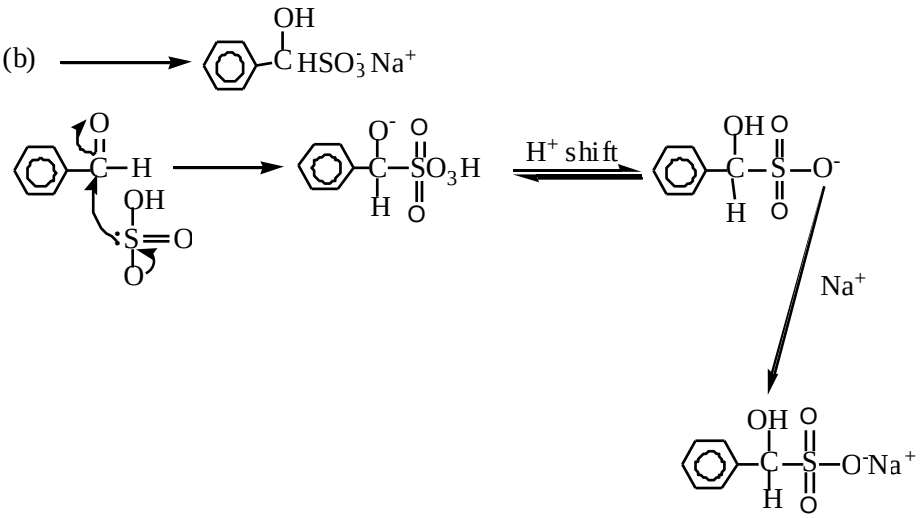


(04 Mks)

$$2H_2SO_4 \rightleftharpoons SO_3 + HSO_4^- + H_3O^+$$

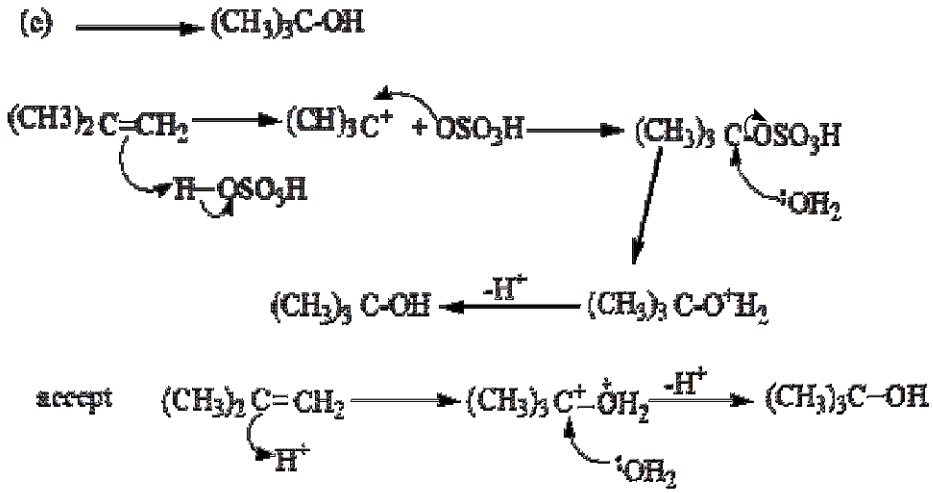


(b)



(03 Mks)

(c)



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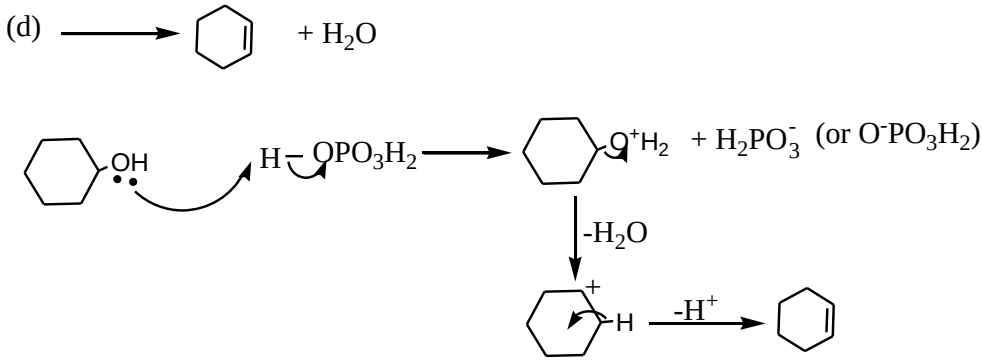
(d)



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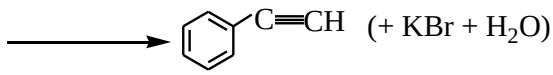
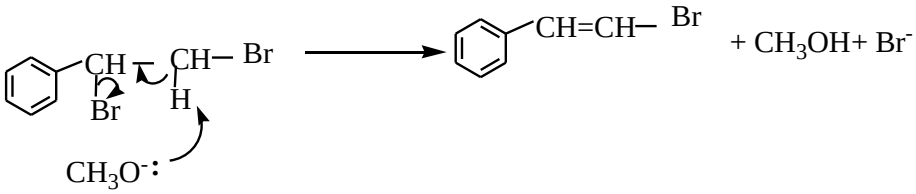
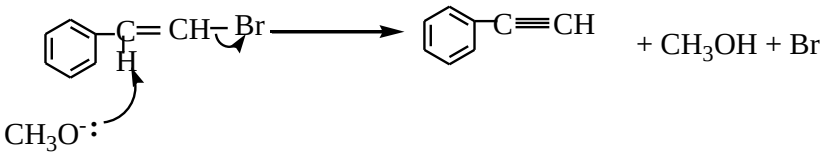
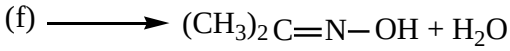
(e) 

(03 Mks)

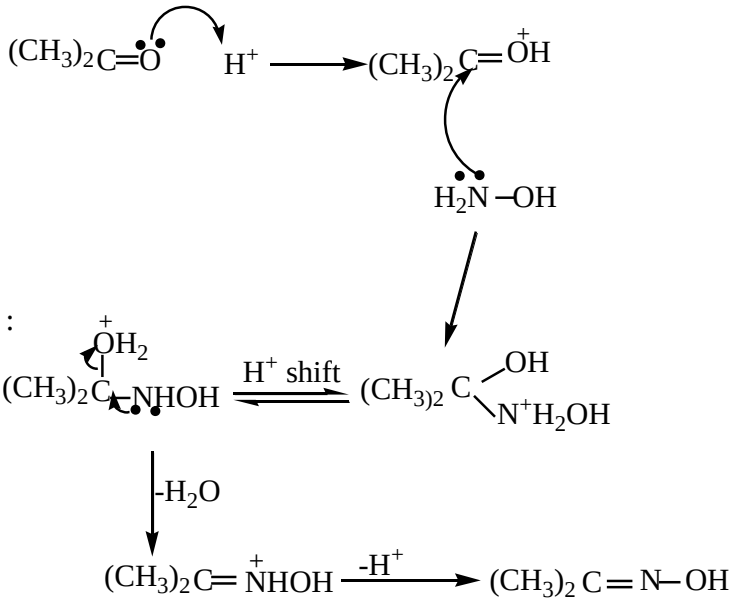
$KOH + CH_3OH \rightleftharpoons K^+ + CH_3O^- + H_2O$





(f) 

(05 Mks)



4. (a) (i) When a solute in shaken with two immiscible solvents in a container (or in contact), at equilibrium the (molar) concentration of the solute in one solvent divided by the (molar) concentration of the solute in the second solvent is a constant at constant (or fixed/particular) temperature. **(02 Mks)**

(ii) -Constant temperature **(01 Mk)**

- Immiscible solvents
- Solute should be soluble in both solvents
- No dissociation/ association of solute in both solvents occurs
- Solute should not saturate any of the solvents
- No chemical reaction between the solvents and solute (*-0½ for extra wrong/correct)*

(b)- A known mass (moles) of butane-1, 4-dioic acid (succinic acid) is added to known equal volumes of water and ethoxyethane in a container/flask. **(06 Mks)**

- The mixture is shaken in a corked container/flash and left to stand at a constant temperature for equilibrium to be established and the layers to separate out.
- A known volume of the organic layer (ethoxyethane) layer is pipette and titrated against a standard solution of sodium hydroxide using phenolphthalein indicator (or a suitable indicator).
- The amount of acid in the organic layer is calculated using the following equation. **(03 Mks)**

$$\begin{array}{l} COOH \\ | \\ CH_{2\,(aq)} \\ | \\ CH_2COOH \end{array} + 2NaOH_{(aq)} \longrightarrow \begin{array}{l} CH_2COONa \\ | \\ CH_2COONa_{(aq)} + 2H_2O_{(l)} \end{array}$$

- The amount of acid in the aqueous layer is obtained by <u>subtraction</u>. The partition coefficient is then calculated from the expression k_d = [acid] in water / [acid] in organic layer.

(c)(i)Let the mass of A extracted be xg **(03 Mks)**

Mass of A left in aqueous layer = $(5.0 - x)$g

k_d = [A] in water/[A] in organic layer

$$k_d = \left(\frac{50 - x}{100}\right) \div \left(\frac{x}{100}\right)$$

(ii) Let the mass of A extracted be yg **(02½ Mks)**

Mass of A left in aqueous layer = (5.0-y) g

$$k_d = \left(\frac{50 - x}{100}\right) \div \left(\frac{x}{100}\right)$$

(d) – A known excess of aqueous ammonia is added to a standard solution of copper (II) ions in a flask/container. **(05½ Mks)**

- An equal volume of an organic solvent (accept named organic solvent) is added to the mixture. The mixture is shaken and allowed to stand at a constant temperature for equilibrium to be established and the layers to separate out.
- A known volume of the organic layer is pipette and titrated with a standard hydrochloric acid solution using methyl-orange indicator (or a suitable indicator).
- The concentration of free ammonia in the organic layer is calculated using the titre values and the following equation of reaction.

  $NH_{3\,(aq)} + HCl_{\,(aq)} \longrightarrow NH_4Cl_{(aq)}$

- From a known value of distribution coefficient, the concentration of free ammonia in the aqueous layer is determined i.e $[NH_3]_{\text{free in aqueous layer}} = K_d[NH_3]_{\text{free in organic layer}}$
- A known volume of the aqueous layer is titrated with a standard hydrochloric acid to determine the concentration of total ammonia.
- By <u>substation,</u> the concentration of complexed ammonia is determined.

  i.e $[NH_3]_{\text{complexed ammonia}} = [NH_3]_{\text{total ammonia}} - [NH_3]_{\text{free ammonia in aqueous layer}}$

- The number of liquids in the complex is determined from the expression.

$$n = \frac{[NH_3]_{\text{complexed ammonia}}}{[Cu^{2+}]}$$

5. (a) (i) $CaF_{2(s)} + H_2SO_{4\,(aq)} \longrightarrow CaSO_{4\,(aq)} + 2HF_{(g)}$ **(1½ Mks)**

(ii) $NaCl_{(s)} + H_2SO_{4\,(aq)} \longrightarrow NaHSO_{4\,(s)} + 2HCl_{(g)}$ **(1½ Mks)**

Allow $H_{2\,(g)} + Cl_{2\,(g)} \longrightarrow 2HCl_{(g)}$

(b) (i) The hydrogen-fluorine bond <u>stronger</u> than the hydrogen-chlorine bond. This is because the fluorine atom has a <u>smaller atomic radius</u> and <u>higher electronegativity than chlorine atom</u>. As a

result fewer hydrogen fluoride molecules ionize in aqueous solution to form hydrogen ions while many molecules of hydrogen chloride ionize in aqueous solution to release hydrogen ions.

(04 Mks)

(ii) Fluorine atom has a smaller atomic radius and is highly electronegative compared to the chlorine atom. The hydrogen-fluorine bond is thus highly polarized compared to hydrogen-chlorine bond. Molecules of hydrogen fluoride associate via stronger hydrogen bonds and molecules of hydrogen chloride associated via weaker Van der Waals forces of attraction. Therefore the boiling point of hydrogen fluoride is higher than that of hydrogen chloride since Van der Waals forces of attraction are weaker than hydrogen bonds. **(03 Mks)**

(c) (i) White precipitated soluble to form a colourless solution **(05 ½ Mks)**

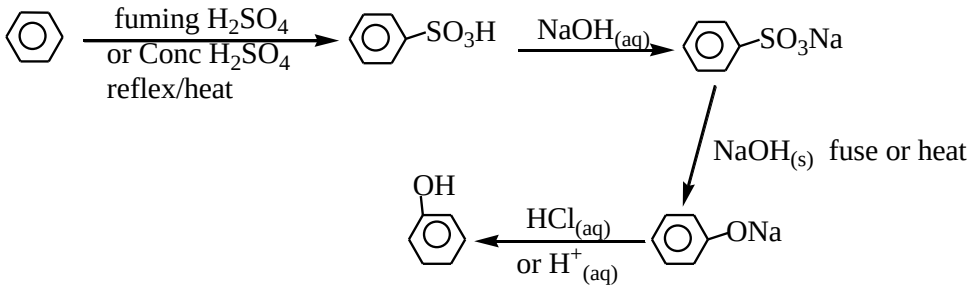
$$Pb^{2+}_{(aq)} + 2Cl^{-}_{(aq)} \longrightarrow PbCl_{2\,(s)}$$

$$PbCl_{2\,(s)} + 2Cl^{-}_{(aq)} \longrightarrow PbCl^{2-}_{4\,(aq)}$$

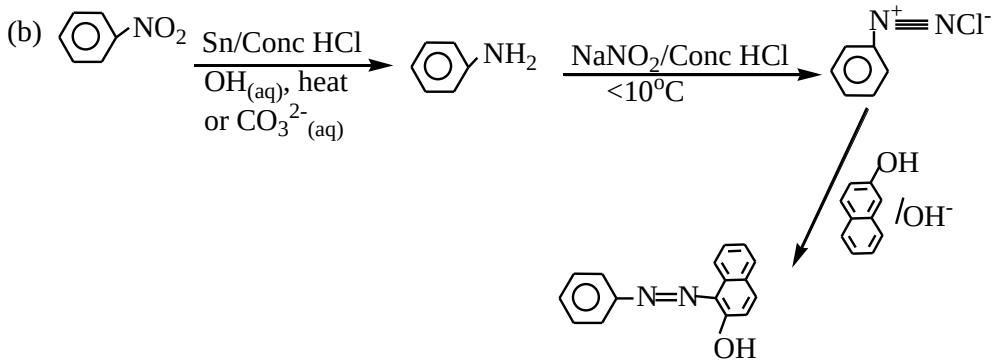
(ii) The blue solution turns yellow **(03 Mks)**

$$Cu(H_2O)^{2+}_{6\,(aq)} + 4Cl^{-}_{4\ 9aq)} \rightleftharpoons CuCl^{2-}_{4\,(aq)} + 6H_2O_{(l)}$$

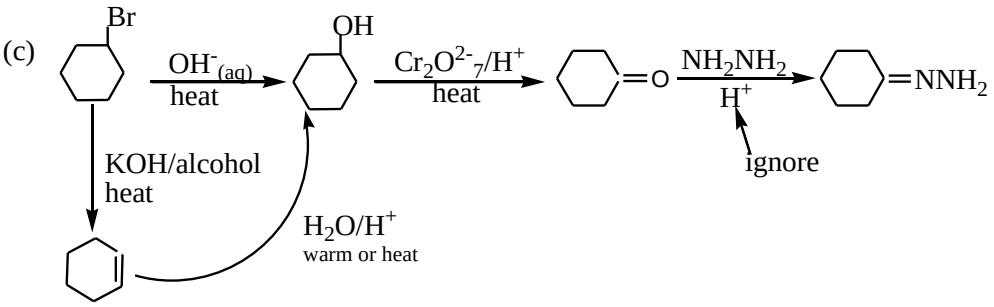
(d) $CuCl^{2-}_{4\,(aq)} + 6H_2O_{(l)} \rightleftharpoons Cu(H_2O)^{2+}_{6\,(aq)} + 4Cl^{-}_{(aq)}$ **(01½ Mks)**

6. (a) 

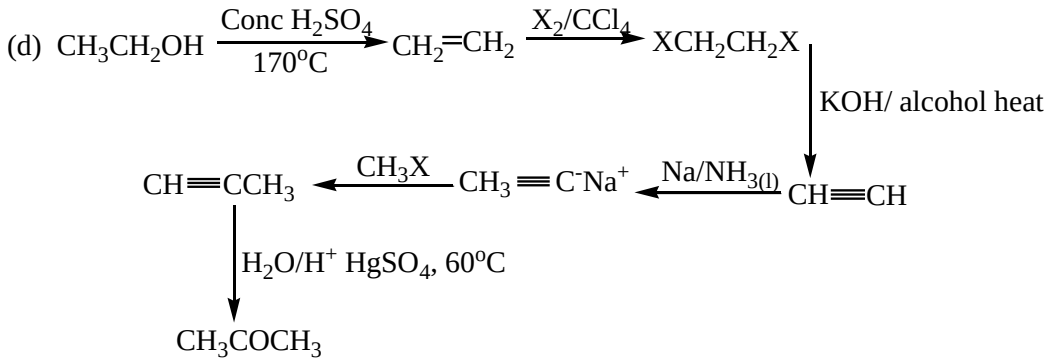
(03 Mks)

(b) 

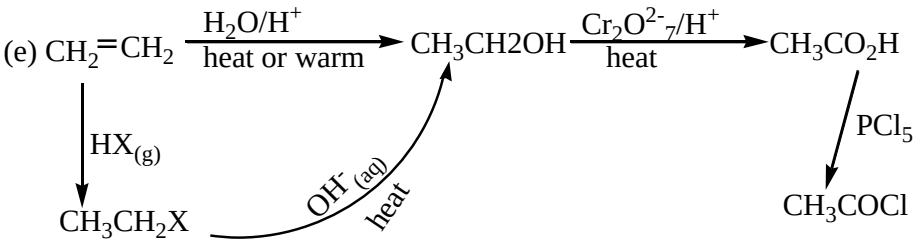
(06 Mks)

(c) 

(03 Mks)

(d) 

(04$^{1}/_{2}$ Mks)

(e) 

(03$^{1}/_{2}$ Mks)

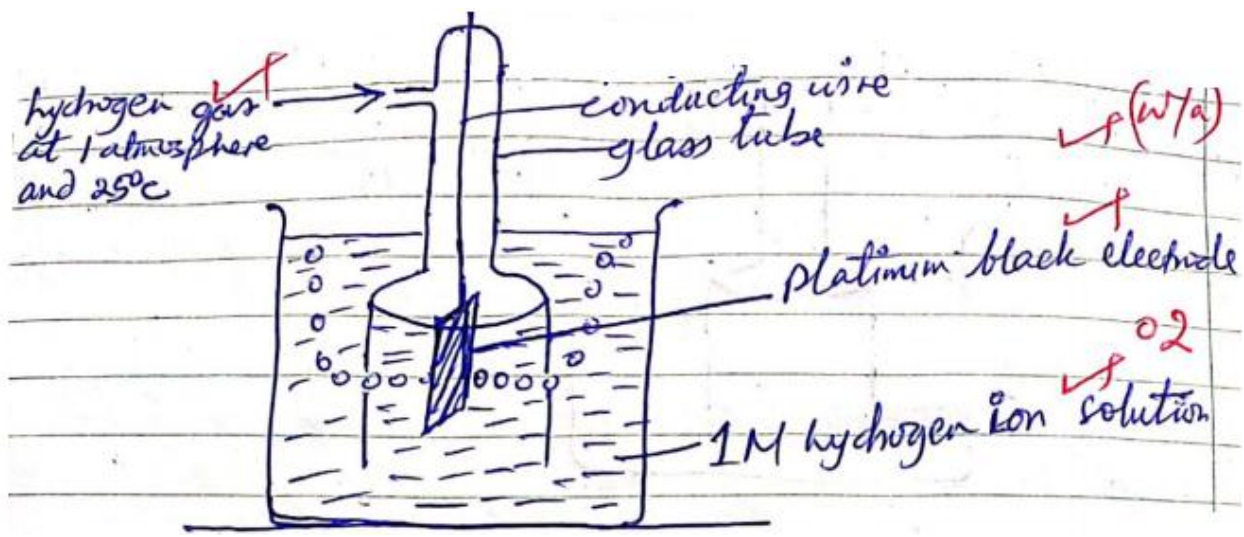
7. (a) (i) Is the reduction electrode potential of an electrode which is the e.m.f of the cell measured with respect to the standard hydrogen electrode at standard conditions. **(01 Mk)**

(ii) Consists of platinum foil coated with platinum black inserted in an aqueous solution containing one molar hydrogen ions the foil acts as inert metal connection. Hydrogen gas at one atmosphere and 25°C is bubbled through the solution.

Hydrogen gas is absorbed on the platinum surface and the following equilibrium is established (between the adsorbed layer of hydrogen molecules and the hydrogen ions) **(04 Mks)**

$$2H^+_{(aq)} + 2e^- \rightleftharpoons H_{2(g)}$$

By IUPAC convention, the electrode potential of the hydrogen electrode is 0.00Volts.



(b)(i) $H_2O_2 + 2H^+_{(aq)} + 2I^-_{(aq)} \longrightarrow 2H_2O_{(l)} + I_{2\,(aq)}$ **(1½ Mks)**

(ii) $Pt_{(s)}/I^-_{(aq)}, I_{2(aq)}//H_2O_{2(aq)}, H^+_{(aq)}, H_2O_{(l)}/Pt_{(s)}$ **(1½ Mks)**

(c) (i) $E^\theta_{cell} = E_{right} - E_{left} = (+1.77) - (+0.54) = +1.23V$ **(02 Mks)**

$$\Delta G^\theta = nE^\theta_{cell}F = -2 \times (+1.23) \times 96500$$

$$= -237390CVmol^{-1}$$

$$= -23739KJmol^{-1}$$

(ii) The reaction is feasible since the ΔG^θ is negative **(02 Mks)**

(d) $Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$ **(03 Mks)**

Quantity of electric current $= 1.75 \times 1 \times 60 \times 60 = 6300C$

1 mole of Cu requires $(2\times 96500)C$

No of moles of Cu deposited $= \frac{6300}{2\times 96500}$

$$\text{Mass of copper deposited} = \frac{6300}{2 \times 96500} \times 63.5 = 2.07g$$

(c) (i) – In an electrolytic cell; oxidation occurs at the positive pole while in the e.m.f cell, oxidation occurs at the negative pole **(02 Mks)**

- In an electrolytic cell, e.m.f is applied to the cell while in an e.m.f, is generated by the cell from the chemical reactions taking place at the electrodes.

(ii) – Extraction of metals **(01 Mk)**

- Purification of metals
- Electroplating

8. (a) –Obtained by electrolysis of concentrated sodium chloride solution (brine) in a cell with a graphite anode and mercury cathode.

- Chlorides ions are discharged at the anode in preference to hydroxyl ions, to form chlorine gas.

  $2Cl^-_{(aq)} \longrightarrow Cl_{2\,(g)} + 2e^-$

- Sodium ions are discharged at the cathode to form sodium ions which <u>dissolve</u> in mercury to form sodium amalgam.

  $Na^-_{(aq)} + e^- \longrightarrow Na_{(s)}$

  $Na_{(s)} + Hg_{(l)} \longrightarrow Na/Hg_{(l)}$ (ignore)

- The amalgam is <u>dissolved</u> in water to form sodium hydroxide and hydrogen. Mercury is also regenerated.

  $Na/Hg_{(l)} + 2H_2O_{(l)} \longrightarrow 2NaOH_{(aq)} + H_{2(g)} + 2Hg_{(l)}$
- The sodium hydrogen pellets is formed by <u>slow evaporation to dryness</u> of sodium hydroxide solution formed.
- Accept use of titamum anode and steel cathode in a cell.

(b) (i) Dilute or concentrated sodium hydroxide solution and <u>heat</u> **(02 Mks)**

$OH^-_{(aq)} + NH^+_{4\,(aq)} \longrightarrow NH_{3(g)} + H_2O_{(l)}$

(ii) <u>Hot concentrated</u> sodium hydroxide <u>solution</u> **(02 Mks)**

$Al_2O_{3(s)} + 2OH^- + 3H_2O_{(l)} \longrightarrow 2Al(OH)^-_{4\,(aq)}$

(iii) <u>Dilute</u> sodium hydroxide <u>solution</u> **(03 Mks)**

$Cr^{3+}_{(aq)} + 3OH^-_{(aq)} \longrightarrow Cr(OH)_{3(s)}$

$Cr(OH)_{3(s)} + OH^-_{(aq)} \longrightarrow Cr(OH)^-_{4\,(aq)}$

(c) Volume of H_2SO_4 for complete reaction with $Na_2CO_3 = 2 \times 10.7 = 21.4cm^3$ **(07 Mks)**

Moles of H_2SO_4 that reacted $= \frac{21.4 \times 0.05}{1000} 1.07 \times 10^{-3}$

$Na_2CO_{3(aq)} + H_2SO_{4(aq)} \longrightarrow Na_2SO_{4\,(aq)} + CO_{2(g)} + H_2O_{(l)}$

Moles of Na_2CO_3 that reacted $= 1.07 \times 10^3$

Moles of Na_2CO_3 that reacted in 1 litre $= \frac{1000 \times 1.07 \times 10^3}{25} = 0.0428$

Rfm of $Na_2CO_3 = (23 \times 2) + 12 + (16 \times 3) = 106$

Mass of Na_2CO_3 in 1 litre $= 106 \times 0.0428 = 4.5368g$

Volume of H_2SO_4 for complete reaction with $NaOH = 23.5 - 10.7 = 12.8cm^3$

$$\text{Moles of } H_2SO_4 \text{ that reacted} = \frac{12.8 \times 0.05}{1000} = 6.4 \times 10^{-4}$$

$$2NaOH_{(aq)} + H_2SO_4 \longrightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$

$$\text{Moles of NaOH that reacted} = 2 \times 6.4 \times 10^{-4} = 0.0512$$

$$\text{Rfm of NaOH} = 23 + 16 + 1 = 40$$

$$\text{Mass of NaOH in 1litre} = 40 \times 0.0512 = 2.048\text{g}$$

$$\text{\% of NaOH in the mxture} = \frac{2.048}{2.048 + 4.5368} \times 100$$

$$= 31.10\%$$